

The Production of Knowledge and Culture: The US 1790-1870*

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Abstract

This paper uses new data on copyright registration title pages from the Library of Congress (LOC) to analyze the intellectual and cultural development of the United States over 1790-1870. I construct national time series of book production over this period which show an uptake in per-capita terms in 1830, well before the start of the Second Industrial Revolution and the era of “knowledge based progress” (Abramovitz & David, 1996). Matching authors to locations (at the county level) using declassified census data reveals that the spatial distribution of intellectual production in the early 19th century is strongly correlated with inventive activity over 1860-1940 and the evolution of the manufacturing sector. Identification is based on a shift-share type instrument exploiting the large internal migration patterns occurring in this time period.

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1 Introduction

According to a popular view in the literature, the principal reason as to why Western countries are richer today than they were two centuries ago is that these countries *know* more (Mokyr, 2005). The sustained accumulation of knowledge is what made technological progress and economic growth possible for today’s developed countries and what separates us from the Malthusian (pre-industrial) world. However, while the proposition that modern economic growth is driven by research efforts leading to useful knowledge is widely accepted in mainstream macroeconomics (e.g. Jones, 2022), we still don’t fully understand when this process began and why. Historians of technology and science emphasize the cultural and political changes which occurred between the 16th and 17th centuries and are sometimes referred to as the Scientific Revolution (Mokyr, 2009). However, the space of papers investigating the relationship between knowledge production and economic development is surprisingly limited (some examples include Baten & Van Zanden, 2008; Chaney, 2016; Dittmar, 2011; Squicciarini & Voigtländer, 2015). This is perhaps partially due to the fact that it is difficult to identify individuals driving knowledge production in historical data, and even if one succeeds in this task, understanding the causal processes at play is far from straightforward. For instance, it may seem evident that sustained innovation requires an expanding body of knowledge, but the amount of resources devoted to knowledge production are likely to be themselves a function of the economic environment and preferences.

In this paper, I use the universe of copyright title pages over 1790-1870, obtained from the Library of Congress (LOC henceforth), to identify the individuals driving *written* knowledge production in the early United States. These primary sources provide an extremely rich set of data concerning the intellectual and cultural history of a nation which, in this time period, went from being a primarily agrarian country to the world’s leading economy.¹ Using census data I map all copyright registrations filed between 1840 and 1860 to the county where the author resided in 1850. This measure of pre-industrial knowledge production is significantly correlated with economic outcomes including population density and the employment share of manufacturing at different points in time. To assuage potential endogeneity concerns, I obtain exogenous variation in the spatial distribution of knowledge producing individuals, through a shift-share instrument which exploits the large amount of internal migration which resulted as a consequence of the Westward movement. Up to 65% of all native individuals who registered a work for copyright between 1840 and 1860 did not reside, in 1850, in the

¹ By the end of the 19th century the US had surpassed Britain to become the world’s largest, richest and most technologically advanced economy (Gallman, 2000).

same state where they were born.² Around 15% of all authors were foreign born, which was considerably higher than the 10% national average³. Using county level shares of individuals born in different states I predict the number of authors residing in a county, exploiting the cross-state differences in copyrighting rates and the fact that in inter-state migration patterns individuals migrate to locations with higher shares of people native to their state of origin. The identifying assumption is that the share of individuals born in each of the 31 US States does not vary systematically with county level unobservables. To mitigate concerns that this assumption may be violated, I control for a wide array of observable socio-economic characteristics and show that these shares do not vary in a systematic way with geographic indicators. Using this strategy, I show that a doubling of pre-industrial copyrighting is associated with an increase in the manufacturing share of employment of around 18 percentage points in 1920, a 78% increase in income per capita in 2000 and a xx% increase in patents per capita filed between 1860 and 1940. This paper adds to the recent literature on the determinants of American long-run structural change and growth which finds that geographic endowments leading to higher agricultural diversity in production in 1860 contributed to the industrialization process (Fiszbein, 2022). I find that, controlling for the abundance of preindustrial human capital identified as the intensity of copyrighting at the county level, there is no effect of agricultural diversity due to better geographic endowments.⁴ This potentially indicates that higher levels of copyrighting activity and knowledge production could be a mechanism through which geographic endowments affected long-run outcomes in US counties.

I then turn to the determinants of copyrighting activity in this period. I find that religious fractionalization in 1850 is the strongest predictor of pre-industrial copyright registrations over 1840-1860. Previous literature has focused on the negative consequences of ethnic and religious diversity in developing countries, highlighting their increased incidence on the severity and frequency of conflicts (Alesina & La Ferrara, 2005; Montalvo & Reynal-Querol, 2003; Montalvo & Reynal-Querol, 2005). In contrast, in the American context increased ethnic diversity during the age of mass migration has been found to have a positive impact on long-run outcomes, potentially due to the importance of skill variety for industrial development (Ager & Brückner, 2013). I construct indices of religious fractionalization and polarization following the methodology of Montalvo and Reynal-Querol (2003) which exhibit, respectively, a positive and negative correlation with intellectual production over 1840-1860. I hypothesize that these correlations reflect an underlying relationship between cultural and religious diversity and the production of knowledge due to the positive effect that competition between different denominations had on the provisioning of religious education. A large portion

² This is the definition of internal migration I use throughout the paper.

³ See Appendix table 18 for more statistics comparing authors to the general population

⁴ See Appendix A.

of copyrighted works consist in textbooks and pedagogical materials for use in *sunday-schools* which were effectively the first schooling institutions in the United States. The lack of data on religious affiliations prior to 1850 makes it difficult to understand the causal mechanisms at play, however the relation between these variables is significant and, to the best of my knowledge, previously undocumented.

The rest of this paper is structured as follows. Section 2 provides a brief overview of the literature. Section 3 describes the data construction process (including the construction of the instrument) and discusses all primary data sources used therein. Section 4 presents some descriptive statistics using the novel data. Section 5 presents the main empirical results concerning the impact of human capital accumulation on regional growth and structural change. Section 6 concludes.

2 Literature Review

This paper is related to a long strand of literature in economic history which points out the role of the “knowledge elite” and their pursuits in stimulating the engine of long run growth (McCloskey, 2008; McCloskey, 2010; Meisenzahl & Mokyr, 2012; Mokyr, 2009, 2016). A key condition which has been emphasized by Mokyr (2016) and Clark (2008) is that the pursuits of this knowledge elite must be sufficiently secular and interested in practical matters. Doepke and Zilibotti (2008) argue that a couple of the richest landowning families in England could have financed the entire capital accumulation necessary for the Industrial Revolution. The reason why the landowning class repeatedly resisted this change instead of capitalizing on it, in their view, was that investment in new technologies and ventures was simply too far removed from their preferences. Using the title pages of books produced by American authors over time period, can therefore help us understand whether the interests of the knowledge elite are becoming more secular and concerned with practical issues, which is indeed what I find in this time period.

Empirical work in this field has generally focused on the more common measures of human capital such as literacy, schooling due to the widespread availability of data in historical censuses. This literature is at least as old as the debate between Acemoglu et al. (2001) and Glaeser et al. (2004) over the “colonial origins” of comparative development. Glaeser et al. (2004) were the first to suggest that heterogeneity in cultural characteristics of colonial settlers could have long-run impacts on the accumulation of human capital and growth, independently of institutional arrangements set up in colonies as argued by Acemoglu et al. (2001). However, given the well-known problems involved in using cross-country regressions to identify causal relationships, these pa-

pers were not able to conclusively determine in what way the international migration associated with colonial settlements affected long-run outcomes. This is a challenging question to answer because, as argued by Glaeser et al. (2004), it remains fundamentally uncertain what Europeans brought with them when they settled to the new world: knowledge, culture or institutions? Most likely a mix of all three and potentially other factors as well. Studies using within country spatial data, where certain features such as the institutional environment can be kept constant, have had more success in establishing a causal link between human capital and economic development. Dittmar (2011) uses city level data from Germany to investigate the effects of the introduction of the printing press, a radical innovation in the technology used to store and produce information, finding large and statistically significant effects. Also in the German context Becker and Woessmann (2009) and Cantoni (2015) find that the protestant reformation had large effects on the accumulation of human capital and, through this channel, city growth and urbanization. In the American context, a growing number of studies has focused on international migration and the skills brought by immigrants when resettling in the United States (Abramitzky et al., 2014; Ferrie, 1999; Sequeira et al., 2020). As noted by Smith (1925), at p.72 “with mercantile sagacity, England prohibited the export of [...] machinery, but she failed to prohibit travel. So one day, Samuel Slater arrived in Providence with a head full of knowledge”. Referred to by Andrew Jackson as the “father of the American industrial revolution”, Samuel Slater was the man responsible for importing British textile technology to the United States. Although Slater’s contributions do not specifically show up in our database due to his lack of written works, it is worth noting that many other key figures in American industrialization, such as Oliver Evans, Thomas Blanchard and John W. Nystrom, who contributed respectively to the advancement of steam power, the assembly line and early computing technology, authored numerous published works, contributing to the dissemination of knowledge in their fields. Identifying the location of these and other individuals who constituted the knowledge elite of the time, as I do in this paper, is thus key to mapping the distribution of human capital and understanding its relationship with the industrialization process.

The closest paper to this article is Squicciarini and Voigtländer (2015), who find that the location of individuals who subscribed to the Encyclopedia in preindustrial France significantly predicted the adoption of technologies and wage growth during the industrial transition of the late 19th century. These authors also found that, consistently with evidence from Britain, conventional indicators of human capital such as literacy rates are only related to economic development in the cross-section, whereas in the time series data only the density of the “upper-tail” of the human capital distribution, in their analysis the encyclopedia subscribers, predicts wage growth and industrialization. By shifting attention to the “upper-tail” of the human capital distribution, being the inventors and knowledge producers, Squicciarini and Voigtländer (2015) reconciled

the robust positive relationship between human capital and economic development in contemporary data with the long-standing belief that conventional indicators such as literacy rates played little role in the industrial revolutions of continental Europe (Mitch, 1993). Contrary to most European countries, the provision of schooling in the US started in a grass-roots fashion at the district level, introduced by local communities of parents and, in some cases, religious organizations (Goldin, 2016). This local endogeneity of schooling provision could therefore provide a channel through which the presence of a knowledge elite, identified from the copyright registrations, promoted economic development.

The interest in books and copyrighted materials as indicators of human capital is, therefore, not necessarily parasitic on the presumed relationship between the production of useful knowledge and productivity growth. Baten and Van Zanden (2008) postulate that book production indicates the presence of a “modern” culture.⁵ Books are themselves more than merely normal goods: their consumption does not only vary with income but also with consumer preferences which are changing as a result of historical processes such as secularization and other cultural changes. The national time-series of book production which I show in Section 5, when compared to the data collected by Baten and Van Zanden (2008), shows the extent of the vertiginous rise of the United States as the human capital leader in the Western world: between 1790 and 1840 the US achieved the increase in per-capita book production attained by European countries over the two centuries leading up to the Industrial Revolution. As in the case of the European countries studied by Baten and Van Zanden (2008), it is important to note that the rise in book production occurred before industrialization began.

3 Data Construction

The data on copyright registrations I use throughout the paper is constructed from the collection of “Early Copyright Title Pages” available from the United States Library of Congress. In order to register an item for copyright in the US from 1790 to 1870, the author had to submit a copy of the title page of the work along with a form and pay a registration fee at their local District Court. This was overhauled in 1870 at which point the responsibility for handling copyright requests was transferred to the Library of Congress along with all the extant copyright records. The nominal registration fee was set at around 60 cents throughout the period under study, which was arguably expensive enough that the author had to believe the work was of some value but not

⁵ See also Mokyr (2016).

so expensive as to be prohibitive to many authors.⁶ Excepting patents, and a small fraction of records which never made it to the Library of Congress, this database arguably comprises the universe of the intellectual production in the US over 1790-1870, or at least the subset which has been preserved in writing. The LOC estimates this corpus to comprise approximately 50 thousand individual copyrighted works in over 90 thousand image records.⁷ To the best of my knowledge, this paper represents the first attempt to classify these records by time, location and subject matter.

The data construction process is done as follows. First, I download the transcriptions of the title pages from the LOC’s website using a webscraper.⁸ I then proceed to process and clean the transcriptions. I extract the year by filtering the transcription for groups of numeric characters starting with “17” or “18”, after converting all roman numerals to western digits. I am able to confidently assign a date to approximately 70 thousand individual entries which is higher than the LOC’s estimate of the total number of works, most likely due to resubmissions and digitization errors by LOC clerks.⁹ I then proceed to extract the name of the author. To do this I simply filter the transcription, removing all words before the first instance of the word “by”, and then match individual words in the transcription to the list of first and last names contained in the 1850 census. A more sophisticated machine learning approach which uses the particular layout of book title pages (which usually contain title, author and publisher name, in that order), following the lines of Shen et al. (2021), would most likely perform more accurately than the procedure I have so far implemented. However, this would require the creation of a large training dataset. The gains to using a layout parser would appear to be limited, I estimate 80% of correct matches, although for works with multiple authors I can only match the first author.

The next step is assigning an author to a location at the county level. To reduce false positives I first extract the name of the state where the book was published, which usually figures on the title page. To correct for misspellings, which are pervasive in historical censuses, I apply *soundex* to both the author names extracted from the primary sources and the individual census records.¹⁰ I then use the first and last name of the author of a work to search for the location based on the census records. In order to have an informative match, I am forced to restrict the pool of potential matches to

⁶ This fee was equivalent to about 20\$ in today’s money. See Khan (2005) for more information.

⁷ The larger number of images is due to some entries being accompanied by forms and blank pages.

⁸ This is perfectly legal as the title pages are in the public domain. All codes are provided upon request.

⁹ Copyright protection expired after 14 years, at which point the author could register their work again for a one time extension. It is important to note that due to human errors there may be multiple entries, that is, the same record may have been scanned twice. Removing the duplicates is, however, not a straightforward process as it is difficult to distinguish them from resubmissions (which could potentially indicate book quality).

¹⁰ The soundex is a user written command available in Stata which assigns to each string a code based on its phonetic pronunciation in the English language, which helps deal with the misspellings which are common in historical census data.

certain high skilled occupational groups (the list of occupations is provided in Table 16). I further impose the restriction that the author must have been at least 25 years old at the time he/she wrote the book. This allows me to exactly match approximately 70% of authors (see the Appendix, Table 15). An exact match here means that for one author I observe one person in the census living in the relevant state with the specified name and satisfying the occupational/age criteria. Unmatched authors are matched in a second round search, which is generally less accurate, by searching for individuals in other states. An important concern here is that any measurement error generated by false positives is likely correlated with population, as finding a false positive is easier in a highly populated area with more potential matches. This generates the potential for spurious results via the construction of a population proxy. To address this concern, I run placebo tests where I randomly select 4000 individuals from the 1850 census and pretend they are authors. Repeating the analysis with these fake authors, reassuringly fails to generate the results obtained for the actual list of authors.

I then classify the works as academic/non-academic. As with most topic classification tasks, there is some arbitrariness in the definition of the topics. I consider a work to have academic value as long as it fulfills two conditions: (i) its main purpose is the diffusion of some kind of knowledge and (ii) this knowledge is not mainly of a religious character.¹¹ An ideal topic classification strategy would use a large training dataset of labeled title pages to classify the corpus (Jelodar et al., 2019). As such a dataset does not exist, I attempt a second best strategy using keywords. Importantly, this will only work for binary classifications (such as academic/non-academic) but not for a more fine grained topic classification.¹² To classify works I construct a list of keywords indicating the presence of scientific knowledge and rank title pages by the relative frequency of these words. This generates a tradeoff between large lists of keywords, which avoid Type 2 errors and smaller lists which avoid Type 1 errors. To solve this I opt for a small list (see Table 20) and use the top 10% of the works and bottom 10%, ranked according to the sum of the relative frequencies of the words in the keyword list, as a training dataset to classify the remaining 80% of the corpus using a logistical regression model of the following form:

$$\mathbf{y} = \mathbf{X}'\beta + \epsilon \quad (1)$$

Where $\mathbf{y}$ is an $N_{key} \times 1$ vector of zeros and ones where N_{key} is the number of works classified using keywords and an entry equal to one means that the title page was in

¹¹ Although some academic works may be written by exponents of the clergy such as manuals for Sunday school teachers.

¹² I also attempted using a Latent Dirichlet Allocation model (LDA) which would allow classification into more fine grained topics without a training dataset. However, the results are disappointing as LDA works best with large amounts of text.

the top 10% of the corpus when ranking by the relative frequency of scientific words, 0 means it was in the bottom 10%. The matrix $\mathbf{X}$ is an $N_{key} \times N_{word}$ where the columns correspond to unique words appearing in the corpus and an entry takes values 0 or n if the word appears in the title page respectively 0 or n times. The $N_{word} \times 1$ vector of parameters β gives the loading of each unique word on the probability that a title page is classified as academic. The error term vector ϵ follows an i.i.d. extreme value distribution. Once we have estimated the coefficients $\hat{\beta}$ we can use this model to predict the topic (academic/non-academic) of the remaining title pages.¹³ This type of classification will work for binary groups such as academic/non-academic. To further distinguish between more fine-grained topics such as religious books, novels, business and accounting publications etc. I use a simple keyword strategy where a title page is put in a category if it contains at least one word on a list of keywords.

4 Descriptive Statistics

This section presents some descriptive evidence relating to the primary sources data constructed in this paper. We start off by examining the topic composition of the entire corpus of works registered between 1790 and 1870 using a simple keyword strategy. The lists of keywords are available upon request, they are not included in the appendix due to space limitations. At this stage we classify a work as belonging to a specific topic as long as it contains at least one of the keywords, which will potentially lead to many overlapping classifications, but is still useful to have a broad overview of the data.

As we can see from Table 1, the keyword strategy for classifying titles leaves a large number of works unclassified and shows substantial overlap between categories. Given the large amounts of overlap in this classification, in the following sections we will be only looking at binary distinctions between academic and non-academic works performed using the logistical regression model described in Section 4. One of the main contributions of this paper is the construction of a national time-series of the number of books registered for copyright in the US over the 19th century, which will then be broken down by location for the 1840-1860 period when most copyrighting occurs. Figure 1 shows the number of works registered for copyright per year over 1790-1870. The large drop in the early 60s is, most likely due to the Civil War.

When normalized by population, we can see that a persistent uptake begins around 1830. This roughly coincides with the earliest growth of manufacturing activity in the US (Davis, 2004), although it is substantially before the introduction of formal

¹³ All codes are available upon request.

Table 1: Descriptive Statistics of Work Categories

Category	Non-Overlap	Overlap	% of Total
Academic	5,389	23,717	9.2%
Textbook	n.a.	9,865	
Invention	n.a.	3,280	
Novel	3,182	10,419	5.4%
Religious	4,486	14,387	7.7%
Business	4,642	16,305	7.9%
Not Classified	26,000	n.a.	44.4%
Ambiguous	n.a.	14,487	24.7%
Total	58,498	n.a.	100%

Descriptive statistics, showing only categories with more than 5000 works classified. The first column shows the number of works falling unambiguously in a category (e.g., scientific works only classified as such), the second shows the total number of works falling in a category allowing for overlap with other categories in the table. Total number of works is defined as the sum of (i) all non-overlapping works in each category, (ii) works falling in more than one category (ambiguous works), and (iii) non-classified works. The third column shows the percentage of each entry in the total. The lists of keywords are available in the Appendix. Textbook and Invention are treated as subcategories of Science and hence do not have a non-overlapping component.

schooling and the massive industrialization which began in the end of the 19th century. Figure 3 plots the number of copyrighted works and patents registered in each decade normalized by the size of the population. As we can see, the rise in copyrighting started around a decade prior to patenting which has been found in previous studies to be substantially related to the growth of markets and a manufacturing industry in the US (Khan & Sokoloff, 1993; Sokoloff, 1988). Regression models in Section 6 will show that the level of knowledge production over 1840-1860 is a significant predictor of patenting during the Second Industrial revolution even when including a wide array of controls and instrumenting for the level of copyrighting exploiting internal migration patterns. Turning to the spatial distribution of knowledge production, Figure 4 shows a map of the US where with state boundaries and copyrighting intensity at the county level. Academic works are shown in blue, whereas non-academic are in red, hence overlap appears as purple. As we can see, while the majority of registrations occur in the Northeast of the country (especially for academic works) there is substantial variation within states, even those in the Midwest and South of the country. While knowledge production is of course non-rival (and only with some difficulty excludable) as emphasized in the endogenous growth literature (Romer, 1990), we may still expect that regions which produce knowledge may vary in certain cultural traits. This can be especially relevant to those works which are not academic and thus speak to the interests and pursuits of the general consumer. In the following section we will turn to how these traits correlated with subsequent economic development. For now, Table 2 shows us the summary statistics of, respectively, above and below median counties ranked by

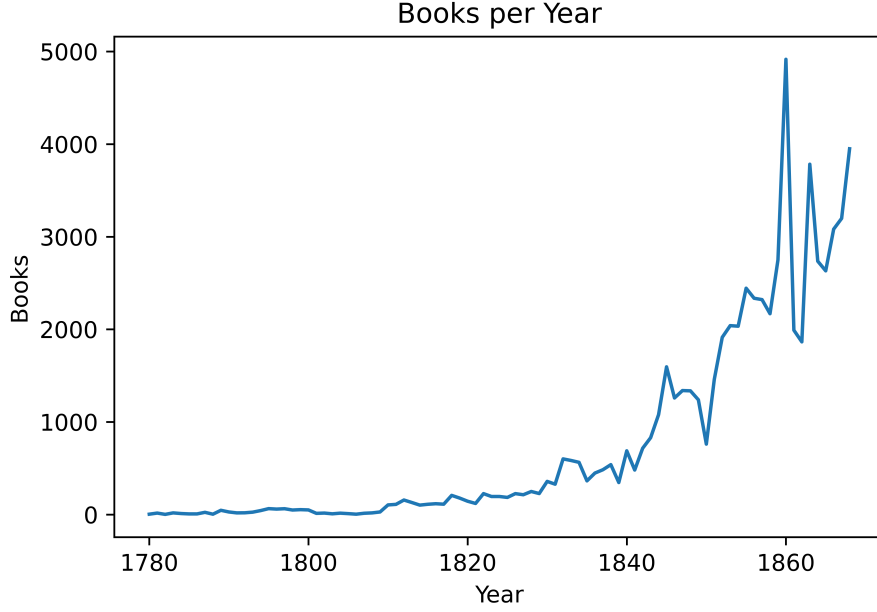


Figure 1: Number of copyrighted works by year over 1790-1870. Source: author's calculations using the Library of Congress (1868) data.

knowledge production. Counties with higher levels of knowledge production exhibit larger populations, higher Solow residuals,¹⁴ literacy and urbanization rates and more colleges and schools (in 1840). In contrast to some of the literature which studies European countries (e.g. Becker & Woessmann, 2009), no particular religious denomination appears to be positively related with copyright registrations. Notice that, while counties with above median levels of knowledge production do indeed have more protestant churches, the levels are exceedingly small (e.g. 3.5 % compared to 2% of all churches). Rather, as I show in Figure 2, indices of religious fractionalization and polarization constructed following Montalvo and Reynal-Querol (2003) respectively correlate positively and negatively with copyrighting activity. These relationships are especially strong (explaining alone around 23% of the variation in copyright registrations) and regression analysis shows that they are significant predictors of copyrighting even when conditioning on a wide variety of controls (see the Appendix Table 17). This is important as the production of knowledge is itself an endogenous product of some historical process and religious competition has been found to have had meaningful consequences for human capital accumulation in Europe by Cantoni et al. (2018). Nonetheless, for now this remains something which is left for future research as a deeper dive into the literature is needed to investigate this point.

¹⁴ The Solow residuals are calculated as OLS residuals from a regression of log output per worker in on capital per worker in the manufacturing sector.

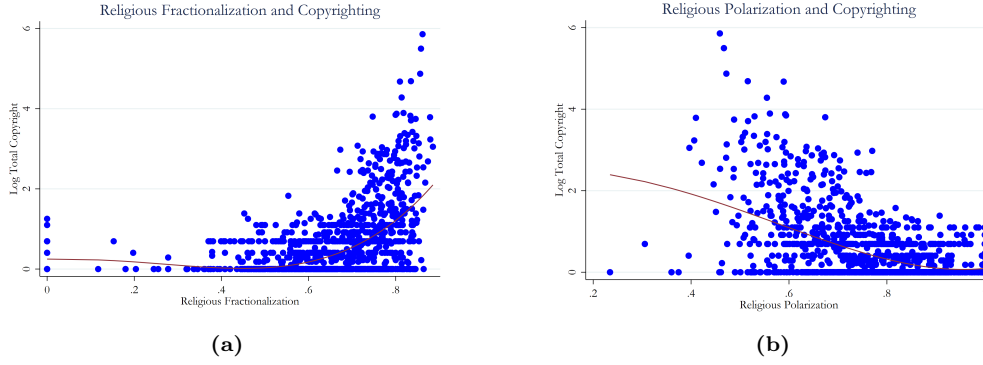


Figure 2: Religious Fractionalization (a) and Polarization (b) against total copyright registrations. For variable definitions and sources see Appendix 9.3 and 9.4.

	Above Median		Below Median		Diff.
	mean	sd	mean	sd	t
Total population	25797.150	33419.058	8367.849	5061.875	(-12.923)
Solow Residual	0.065	0.403	-0.074	0.501	(-5.168)
Literacy %	0.876	0.114	0.792	0.151	(-11.134)
Urbanization %	0.021	0.120	0.000	0.000	(-4.284)
#Slaves/#Population	0.167	0.223	0.211	0.226	(3.433)
# Colleges	0.306	0.720	0.053	0.230	(-8.390)
# Schools per child	0.020	0.014	0.015	0.012	(-5.605)
Protestant churches %	3.520	8.308	2.056	6.292	(-3.212)
Catholic churches %	3.804	8.144	5.014	15.123	(1.733)
German Reformed churches %	1.349	4.055	0.395	1.982	(-3.765)
Observations	628		629		1257

Table 2: County-level summary statistics by intensity of copyrighting. Columns (1) and (2) show summary statistics for respectively above and below median counties in terms of total registered works for copyright. Sources: data on copyrighting is based on author's calculations using Library of Congress, 1868, all other data is from Haines et al., 2010.

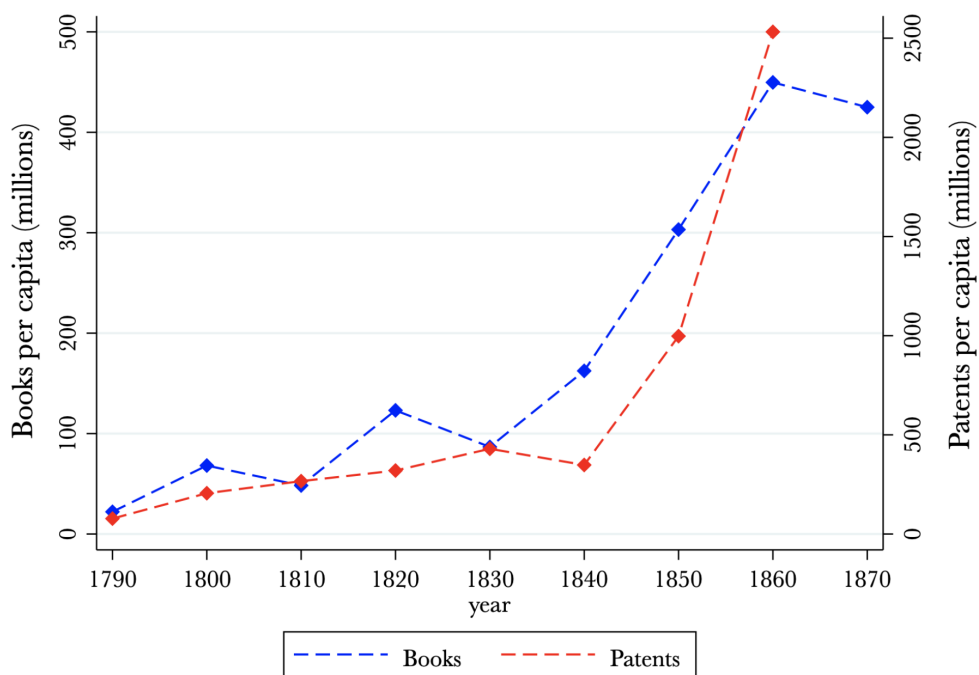


Figure 3: Copyrighted works and patents normalized by population. Source for copyright data: author's calculations using the Library of Congress (1868). Source for patents: USPTO (2022).

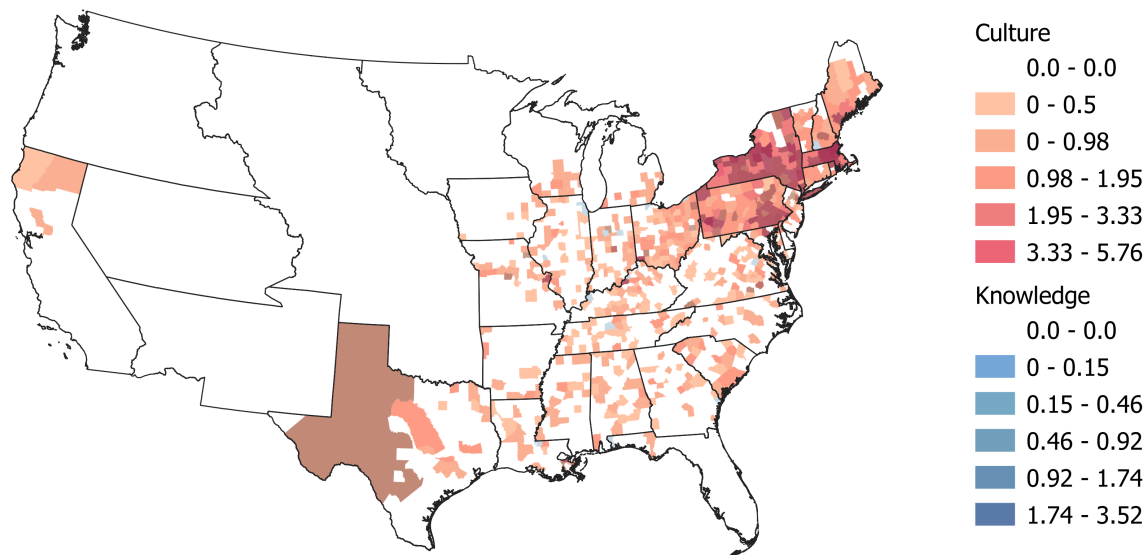


Figure 4: Log of copyrighted works over 1840-1860 by county. Academic and non-academic works are respectively in blue and red, hence overlap appears as purple. Source: author's calculations using Library of Congress (1868).

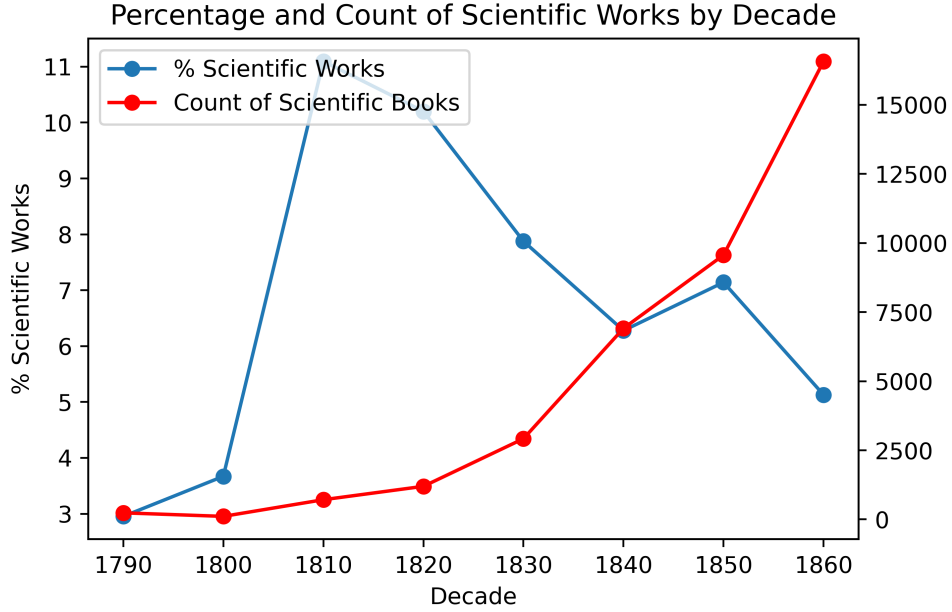


Figure 5: Total and relative frequency of works classified as “scientific” following the methodology laid out in Section 3. The code used to classify works is provided in the Appendix. Notice that these results need not be consistent with those presented in Table 1 as they come from a different classification procedure.

Turning to the topic composition of copyrighted works, as we can see in Figure 6, the total count and fraction of all works which can be classified as academic is increasing over time reaching a peak of 22%. This would seem to indicate substantial interest in the pursuit of secular knowledge starting long before the rapid industrialization experienced in the second half of the 19th century. If we restrict our attention to works which have scientific value, by which we mean works attempting to push the knowledge frontier forward, these appear to be declining in relative importance from 1810 onward.¹⁵ This is not surprising as the growth of the reading public implies that more accessible publications are growing in demand faster than scientific ones.¹⁶ To better understand the topic composition of the growing intellectual production we observe in this time period we can use a wordcloud to visualize the most frequent words by decade. Although purely descriptive in nature, such an exercise can be particularly useful when a time dimension is included as it can show whether the focus of the knowledge elite is changing or stable over time. Table 3 below shows wordclouds over 20 year intervals for the whole sample period.

¹⁵ This classification is done following the same methodology outlined in Section 4 for the academic/non-academic classification.

¹⁶ Being literate in 1790 is far less common than being literate in 1870. As the reading public grows it is natural to expect that the focus of publications should shift towards layman’s topics.

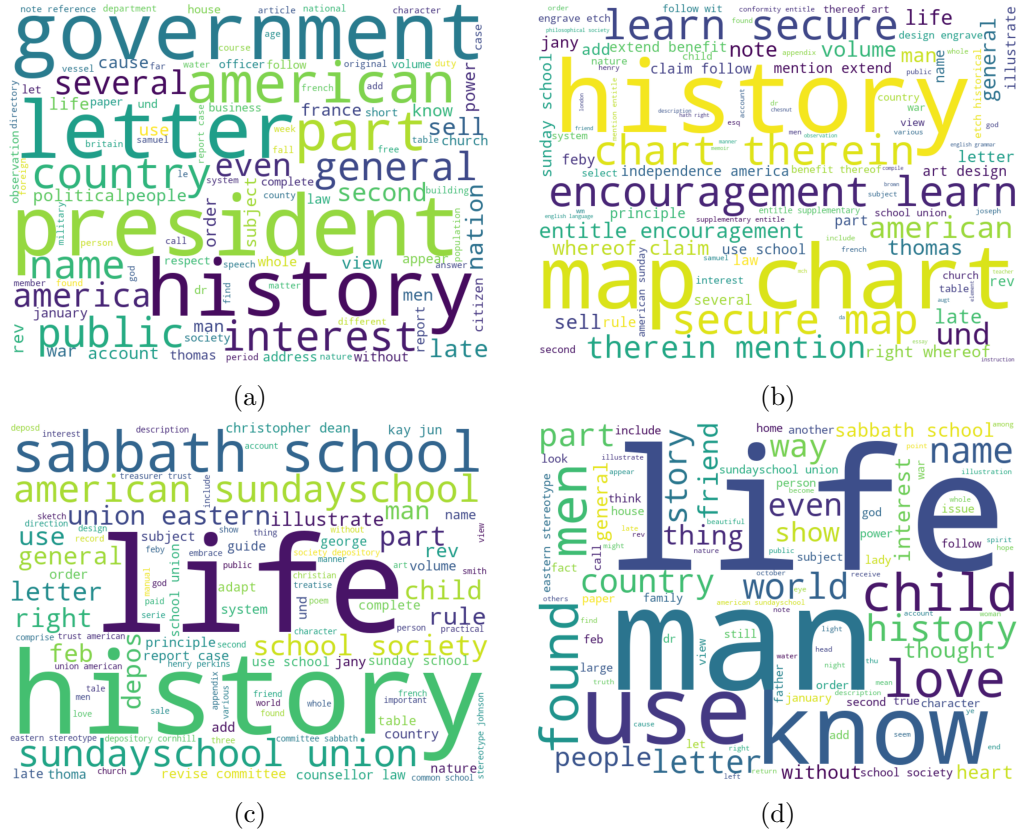


Table 3: Wordcloud plots of the corpus at different moments in time. Panels (a), (b), (c), and (d) refer respectively to time periods 1790-1810, 1810-1830, 1830-1850 and 1850-1870. The list of excluded stopwords is available in the code provided in the Appendix. The size of each word in the cloud is proportional to its frequency in the corpus of text.

The early focus of intellectual production in the US appears to be concerned with books relating to American history and accounts of the parliamentary discussions and political discourse of the time. However, as early as the period between 1810 and 1830, we can see that the subject matter changes noticeably. An increasing importance of the words “learn”, “encouragement” and, in the following periods, “school” and “know” indicates that the spread of education in the US may have started well before the introduction of mass schooling towards the end of the 19th century. The appearance of the word “life” in the period after 1830 is itself interesting and calls for further investigation. While this could be related to the “life sciences”, this is unlikely to be the case as the word “science” appears nowhere in these wordclouds. Most likely, the frequency of this word is the result of the growing number of people who feel the need to write their autobiography or the biography of key figures in American history. This is an interesting result as it indicates that people feel their lives, and those of the key figures in the history of their country, are worth recording and transmitting to future generations. This is highly significant as a marker of economic change as traditional societies where life is “nasty brutish and short” are, arguably, unlikely to be characterized by this kind of behavior.

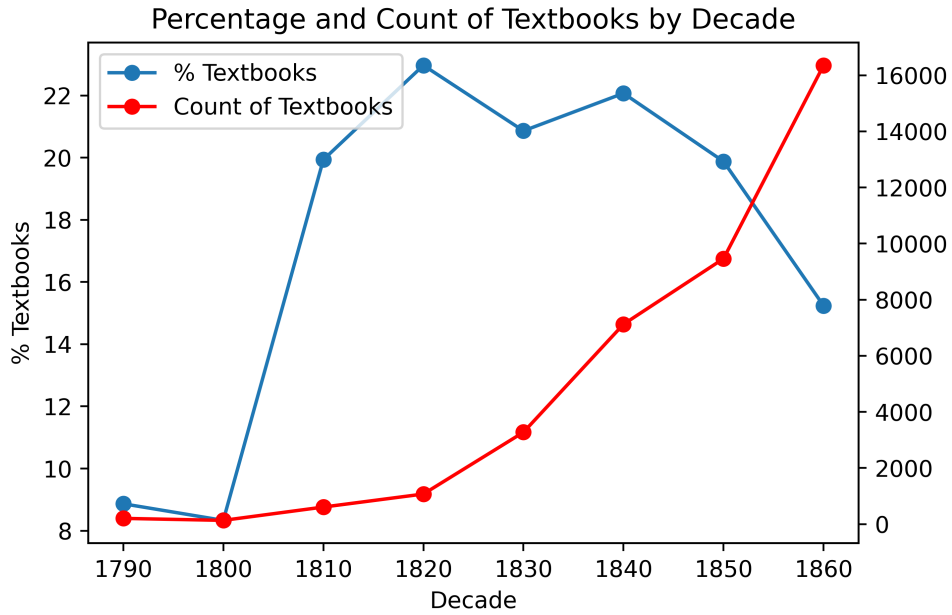


Figure 6: Total and relative frequency of works classified as “Textbooks” following the methodology laid out in Section 3. The code and list of keywords used to classify works is provided in the Appendix. Notice that these results need not be consistent with those presented in Table 1 as they come from a different classification procedure.

5 Empirical Methods

5.1 Instrument Construction

As discussed in Section 3, I will use an empirical strategy which leverages the large patterns of internal migration occurring in this time period. Table 4 shows the fraction of individuals by region who are classified as internal migrants in the 1850 census.¹⁷ As we can see, this fraction is especially high for the Midwestern region and is always higher for authors, although it is still substantial even in the Northeastern region which was already largely settled by the end of the 18th century. The identification strategy exploits the fact that authors, just like other migrants, tend to choose locations where larger fractions of people from their home state settled. This is what is known in the migration literature as the “migrant network instrument” (Bartel, 1989; Munshi, 2003), with the exception that I have no time variation as data on state of birth only becomes available in the 1850 census.¹⁸

Region	Mean		Frequency
	Non-Author	Author	
Midwest	0.45	0.78	4,584,383
Northeast	0.12	0.30	7,207,901
South	0.22	0.48	5,434,905
Total	0.24	0.36	17,227,189

Table 4: Summary Statistics of Internal Migration for natives by Region and Author Status. Source: US Census of Population 1850 and Copyright Registration Data from the Library of Congress.

Nonetheless, I argue that this instrument will alleviate selection bias concerns as factors which induce the knowledge elite to settle in particular areas are likely to cut across state origins: that is, they attract authors from Connecticut just as they do authors from New York. This could be important as, if authors across states tended to choose particular locations with better unobservable characteristics OLS coefficients may capture a selection effect, due to authors systematically choosing better locations, as opposed to an effect of knowledge production on subsequent economic development. Alternatively, measurement errors in knowledge production due to imperfect matching and failure in identifying author names from the primary sources will almost certainly lead to downward bias in OLS coefficients. The instrument will thus exploit the fact

¹⁷ As discussed above, my definition of internal migrant is an individual who did not live in 1850 in the same state where they were born.

¹⁸ The traditional “migrant network instrument” uses shares from the previous period (which are considered to be more plausibly exogenous) to predict current migration flows. In a traditional migration paper using the current shares would be inappropriate as there would be a, meaningless, mechanical relationship between the level of immigrants from a certain state and their share, however in our case this is not true as authors are a very small subset of the population (approximately 4000 out of 19 million).

that authors tend to locate in regions with larger shares of individuals from their state, alleviating the selection concern, and that individuals from states with higher levels of copyrighting are more likely themselves to produce more unobservable knowledge, alleviating the measurement error concern.¹⁹ The identification assumption is that, aside from having higher levels of human capital, settlers from different states are otherwise similar and they do not select systematically different locations. While this assumption is admittedly strong, especially since other characteristics of Westward moving settlers have been studied in the literature (Bazzi et al., 2022; Bazzi et al., 2021), we can control for a wide variety of county level characteristics to identify other channels through which initial settlement patterns may affect subsequent development. Formally the instrument is constructed as follows:

$$\hat{books}_{i,s} = \sum_{l \neq s} \left(\frac{authors_l}{people_l} - \frac{authors_{us}}{people_{us}} \right) \pi_{i,l} \quad (2)$$

Where the predicted level of copyrighting in county i of state s is a weighted sum of the average deviations (from the national average) of copyrighting per capita in all other states l ($authors_l$ is the number of people who registered a work for copyright between 1840-1860 and were born in state l and $people_l$ is the number of people in the 1850 census born in state l) weighted by the relative fraction of individuals from that state in the county population (in 1850). I also add the largest foreign suppliers of international migrants to the list of US states. These countries are the UK, France and Germany, which are treated as if they were US states in 2.²⁰ This prediction is used as an instrument for the actual level of copyrighting at the county level to isolate the part of knowledge production which is due to the cultural composition of settlers and not based on selection mechanisms which would attract authors across states. To motivate this as a valid instrument, we can run a reduced form regression to see how the instrument correlates with the observable geographic characteristics of different counties. If there is selection on observables (meaning that people from states with high levels of copyrighting choose better locations), there may be also selection on unobservables which casts doubt on the validity of the instrument. To test this, consider the following model:

$$\hat{books}_{i,s} = \alpha_s + \delta \Gamma_{i,s} + \epsilon_{i,s} \quad (3)$$

Where α_s is a state fixed effect and Γ is a vector of geographic controls including average temperature and precipitation, distance to lakes and oceans, potential yields

¹⁹ In the Appendix I formalize this conjecture with a Monte Carlo experiment. See Figure 12.

²⁰ As we have no data on copyright registrations outside the US, the fraction $authors_l/people_l$ is the rate of copyrighting for individuals from those countries in the US.

and terrain ruggedness (measured as the average slope in a county). The results are shown in Table 5 which shows that, if anything there is negative selection. People from states with higher levels of copyrighting activity tend to resettle in areas which are farther from oceans and lakes, more rugged and colder. The association with land suitability and mean precipitations is insignificant.

Table 5: OLS Regressions: Selection on Observables

	(1)	(2)	(3)
Terrain Ruggedness	0.811*** (3.79)	0.848** (3.18)	0.581** (3.19)
Distance to the Ocean or the Great Lakes (in km)	-0.0367*** (-9.03)	-0.0333*** (-5.55)	-0.0102 (-1.43)
Temperature	-1.069** (-3.22)	0.947 (1.71)	1.514 (1.58)
Annual Rainfall	0.000104 (0.02)	-0.00642 (-0.81)	0.00652 (1.24)
Land Suitability	-1.744 (-0.50)	4.738 (1.73)	4.760 (1.89)
N	1507	1507	1507
R2	0.423	0.145	0.235
State FE	no	yes	yes
Lat/Lon Polynomial	no	no	yes

t statistics in parentheses.

Dependent variable is the instrument constructed using 3. Conley standard errors used with a 100km cutoff. For variable definitions and sources, see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5.2 Main Results

We now turn to the relationship between pre-industrial human capital and economic development over the course of the Second Industrial Revolution and into the 20th century. The basic empirical model we are interested is the same one Fiszbein (2022) which takes the following form:

$$y_{i,t} = \alpha_s + \beta \ln(\text{Copyright}_{1840-1860}) + \Gamma' X_{1850} + \varepsilon_{i,t} \quad (4)$$

Where $y_{i,t}$ is an outcome variable measured at time t , α_s is a state fixed effect, $\ln(\text{Copyright}_{1840-1860})$ is the variable of interest which is the logged number of copy-

right registrations between 1840 and 1860 in a given county²¹ and X is a vector of control variables. There are three types of controls considered: geographic controls which are unaffected by human interaction with the environment (these are potential yields, average rainfall and temperature, distance to oceans and the great lakes and terrain ruggedness), social controls which are potentially endogenous (these are log of population, urbanization, the fraction of foreigners and internal migrants and population density) and human capital controls which account for the potential correlation between knowledge production and other types of human capital which could have independent effects on economic development (these are literacy, and school enrollment rates). All social and human capital controls are measured in 1850. In addition to these controls, all regressions (unless stated otherwise) include a third degree polynomial of longitude and latitude to limit concerns relating to spatial autocorrelation (e.g. Kelly, 2020).²² Furthermore, standard errors are calculated using the Conley correction (Conley, 1999) with a cutoff of 100km in all tables. This means we allow for dependence between observations within 100km for each other. Alternative estimates using state clustered standard errors are qualitatively equivalent. One advantage of this specification compared to ?? is that by keeping the variable of interest fixed in time, we can limit reverse causality concerns running from the dependent variable to the variable of interest. This is also a similar empirical setup to Squicciarini and Voigtländer (2015), Fiszbein (2022) and several other papers which study path dependence and persistence in economic outcomes.

The first outcome variable we consider is city growth in during the Second Industrial Revolution (1860-1940). Following a long tradition in the historical political economy literature, population growth - and urban population growth in particular - is often considered to be a fairly decent proxy of economic development (De Long & Shleifer, 1993). The presence of knowledge elites has already been found to predict city growth during industrialization in France by Squicciarini and Voigtländer (2015), who stress that while traditional measures of human capital such as literacy rates correlate with economic development in the cross-section, only upper-tail human capital predicts changes in the long-run. The results I show in Figure 7 and Tables 6 and 7 are consistent with these findings, showing that the intensity of copyrighting activity over 1840-1860 strongly predicts the growth in the share of people living in urban centers.

We now turn to the labor share of manufacturing and population density over 1850-2020, to understand the mechanisms through which pre-industrial knowledge elites influenced the development trajectories of counties. While population density is a general proxy for economic development both in pre-industrial and modern times, the

²¹ The period 1840-1860 is chosen as the only census which I have available to use outside of the Minnesota Population Center’s remote desktop computers is the 1850 one. In future versions of this paper this variable should be the total over 1790-1850. However, given that copyright registrations only pick up speed after 1830 this should change the results by much.

²² The results are not sensitive to changing the degree of the polynomial.

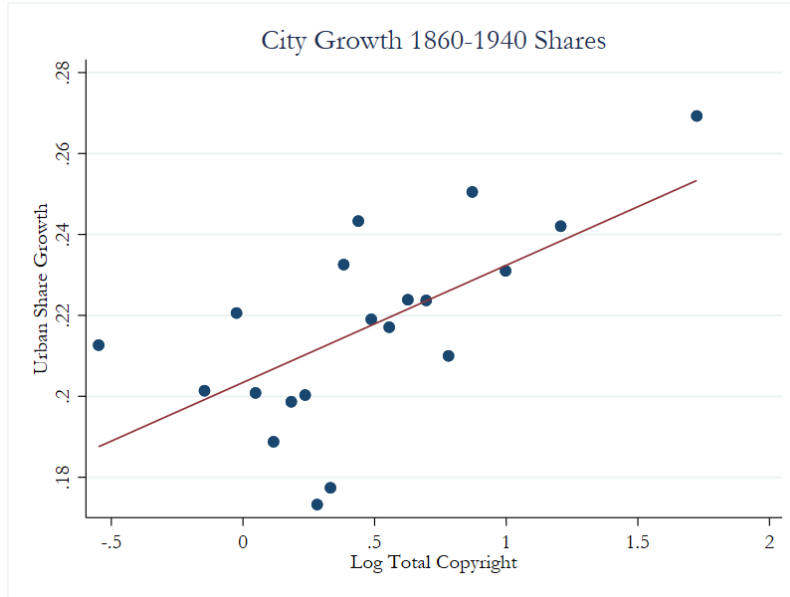


Figure 7: Binscatter plot of the growth of urban share population over 1860-1940 against the log of copyright activity between 1840-1860. Number of quantiles is 20 and all controls - social, human capital, geographic, the latitude-longitude polynomial and state fixed effects - are partialled out. For variable definitions and sources see the Data Appendix.

labor share emphasizes the relationship between human capital and industrialization which is at the core the economic transformations occurring in the US and other Western countries in the end of the 19th century. To motivate this relationship I will show that copyrighting is a significant predictor of innovation as measured by per-capita patent rates over 1860-1940. Surprisingly, the variable of interest appears to be only weakly related to the Solow residual over 1850-1920.²³ This, however, may be due to heterogeneity in prices between counties as output per worker is measured in nominal terms. As data on county level prices is unavailable, I do not show these results here.

As we can see in Table 8, even in the most conservative specifications OLS estimates show a significant relationship between knowledge production in the pre-industrial period and structural change. In all tables, the coefficients shown are a selection of the complete set of 20 control variables included in the fully specified models. The point estimates in Table 8 peak around 1920 and subsequently decline, following the hump-shaped pattern of manufacturing shares exhibited by most developed countries over this time period. Controls are often insignificant (e.g. urbanization) or enter the regression with the wrong sign (e.g. population density and schooling enrollment rates). No variable in 4 predicts the structural change pattern as well as the log of

²³ The Solow residual is calculated as an OLS residual of a regression of log output per worker on log capital per worker in manufacturing.

Table 6: IV Regressions: Urbanization Share Growth 1860-1940

	(1)	(2)	(3)	(4)	(5)
Log Copyright	0.328* (2.38)	0.365* (2.45)	0.358* (2.13)	0.358* (2.13)	-2.832 (-0.05)
% Urbanized 1850	-1.200*** (-3.44)	-1.279*** (-3.62)	-1.260** (-3.12)	-1.260** (-3.12)	1.954 (0.04)
% In School 1850			0.00831 (0.23)	0.00831 (0.23)	-0.127 (-0.04)
% Literate 1850			0.0678 (0.26)	0.0678 (0.26)	-0.698 (-0.04)
Log Population 1850					0.561 (0.06)
N	1472	1467	1466	1466	1465
Kleibergen-Paap F	16	17	13	13	1
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is city population growth in levels over 1860-1940. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

copyright registrations over 1840-1860. While these results suggest the presence of a relationship between pre-industrial knowledge elite presence and industrialization, we may still be capturing spurious selection effects. As we are controlling for a significant number of observable characteristics of counties, including several controls which may be potentially endogenous to the variable of interest, we can interpret the estimates in Table 8 as being fairly conservative. Nonetheless, counties with higher levels of knowledge production in pre-industrial periods may be different in terms of unobservable characteristics. The main concern is that, given the high rates of internal migration, authors and other knowledge producers select locations which have better chances of industrializing at the end of the 19th century. This is partially addressed by controlling for urbanization and population density in 1850, however, these variables may still imperfectly capture the factors driving selection and hence only partially address the problem.

An instrumental variable strategy is therefore needed to isolate the effect of knowledge production on industrialization which is not driven by pre-existing characteristics operating through selection mechanisms. Table 9 addresses this by using the shift-share level of predicted knowledge production described in Section 4. This instrument ex-

Table 7: OLS Regressions: Urbanization Share Growth 1860-1940.

	(1)	(2)	(3)	(4)	(5)
Log Copyright	0.0654*** (4.45)	0.0617*** (4.36)	0.0548*** (3.92)	0.0548*** (3.92)	0.0169 (1.11)
% Urbanized 1850	-0.503*** (-7.98)	-0.504*** (-7.66)	-0.482*** (-7.51)	-0.482*** (-7.51)	-0.712*** (-8.02)
% In School 1850			0.0673** (2.98)	0.0673** (2.98)	0.0285 (1.31)
% Literate 1850			0.386* (2.20)	0.386* (2.20)	0.202 (1.16)
Log Population 1850					0.0431*** (3.37)
N	1472	1467	1466	1466	1465
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is city population growth in levels over 1860-1940. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

exploits the fact that when migrating authors tend to choose locations which have relatively larger settlements of individuals from their home state. Identification here comes from the fact that any factor which is potentially attractive to authors, and hence introduces selection bias in our coefficients, is likely to cut across state boundaries. It is challenging to think of a selection mechanism which operates only for individuals from Connecticut and not for those from New York and simultaneously affects long-term development trajectories. It may however be true that individuals from Connecticut are different from individuals from New York in dimensions which go beyond the relative abundance of authors in New York and relative scarcity in Connecticut. To address this we keep controlling for all the human capital controls and urbanization in 1850. We are however forced to drop the rest of the social controls as partialling them out in the first stage leads to a weak instrument problem which generates imprecise coefficients (although the sign of point estimates is stable). The controls which are driving this problem are the fraction of internal migrants and foreigners, which is unsurprising given that they have an explicit relationship with the instrument.

As we can see from Table 9, point estimates are larger in the IV specification and broadly follow the same pattern. The instrument is strongly relevant as can be seen

Table 8: OLS Regressions: Labor Share all Controls

	(1) 1850	(2) 1870	(3) 1920	(4) 1970	(5) 2000
Log Copyright	0.00742*** (5.65)	0.00927*** (3.57)	0.0111*** (3.50)	-0.00324 (-1.06)	-0.00239 (-0.51)
% In School 1850	-0.0104** (-2.85)	0.00651 (0.57)	0.0160** (2.69)	0.0304*** (4.38)	0.00853 (1.13)
% Literate	-0.0114 (-0.65)	0.0556 (1.18)	0.0601 (1.58)	0.0460 (1.73)	0.0241 (0.67)
% Urbanized	0.0218 (1.06)	-0.00409 (-0.18)	-0.0814*** (-3.51)	-0.0602** (-2.63)	-0.00836 (-0.47)
Log Population 1850	0.000840 (0.34)	-0.00419 (-0.66)	0.00127 (1.09)	0.00103 (0.48)	0.00939** (2.82)
N	1114	1504	1504	1504	1504
R2	0.271	0.193	0.204	0.221	0.129
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	yes	yes	yes	yes	yes
HC Controls	yes	yes	yes	yes	yes
Social Controls	yes	yes	yes	yes	yes

t statistics in parentheses

Dependent variable is the labor share of manufacturing 1850-2000. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

from the large Kleibergen-Paap statistics which are generally above 25.²⁴ The human capital and urbanization controls enter the regression with the wrong sign and are generally not significant. It is important to comment on the relative magnitudes of the IV and OLS coefficients as large differences between these parameter estimates can sometimes indicate issues of instrument endogeneity (Oster, 2019). Selection on unobservables would if anything lead to upward bias in the OLS coefficients, which is exactly the opposite of what we observe. Hence, the likely source of downward bias in the OLS coefficients is due to measurement error in the variable of interest. This measurement error has two sources. Firstly, the variable we are measuring (total copyrighting at the county level) is an imperfect proxy for the general level of knowledge production. While a large component of this is likely to be unobservable in copyright records, the instrument can potentially correct for this by considering the variation in copyrighting activity by states. As long as individuals from states which produce more

²⁴ However, if we add the rest of the social controls (which are population and the share of foreigners and internal migrants) the association between the instrument and the variable of interest vanishes leading to large standard errors.

Table 9: IV Regressions: Labor Share all Controls

	(1) 1850	(2) 1870	(3) 1920	(4) 1970	(5) 2000
Log Copyright	0.0630*** (16.47)	0.0360 (0.44)	0.185*** (3.88)	0.102 (1.96)	0.0350 (0.81)
% In School 1850	-0.0234*** (-5.53)	0.00458 (0.19)	-0.00765 (-0.73)	0.0174 (1.54)	0.0101 (1.15)
% Literate	-0.0520** (-3.05)	0.0328 (0.28)	-0.102 (-1.38)	-0.0548 (-0.70)	-0.00929 (-0.12)
% Urbanized	-0.0597* (-2.15)	-0.0119 (-0.06)	-0.401*** (-3.42)	-0.253 (-1.83)	-0.0781 (-0.68)
N	1114	1505	1505	1505	1505
Kleibergen-Paap	23	13	13	13	13
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	yes	yes	yes	yes	yes
HC Controls	yes	yes	yes	yes	yes
Social Controls	no	no	no	no	no

t statistics in parentheses

Dependent variable is the labor share of manufacturing 1850-2000. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

copyrighting also produce, on average, more unobservable knowledge the instrument can address this issue. Secondly, the measurement of copyrighting activity by county is itself subject to error due to potential false matches and failed extraction of author names from the title pages. Monte-Carlo experiments can show us that these two sources of measurement error can generate substantial levels of downward bias in OLS coefficients (see Appendix Figures 12a and 12b).

To better visualize the strong relationship between pre-industrial human capital and structural change, Figure 8 shows a coefficient plot of the variable of interest over time using the preferred specification, which includes all except social controls (but including urbanization). We can see that, while there are large differences between the OLS and IV coefficients, these estimates follow the same patterns and adding more years does not overturn the results in Tables 8 and 9. Figure 9 applies the same procedure to population density, showing a robust and increasing positive association with pre-industrial knowledge production. Agricultural diversity, which was argued to have exerted a determining role in the US' structural change in a recent paper (Fiszbein, 2022), appears to have no explanatory power once pre-industrial human capital measured through copyrighting activity is taken into account (see Figure 11 in the Appendix).

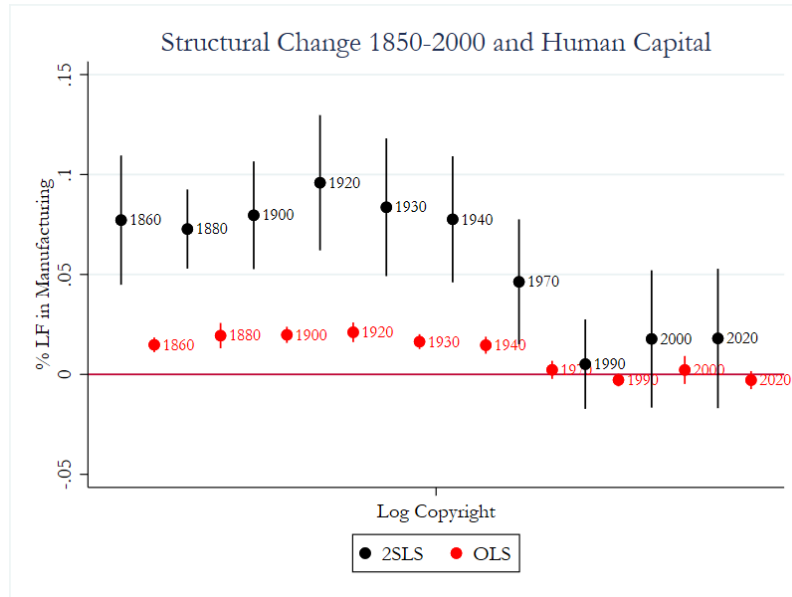


Figure 8: Point estimates with 95% confidence intervals of the effect of copyrighting in 1840-1860 on the labor share of manufacturing over 1850-2020.

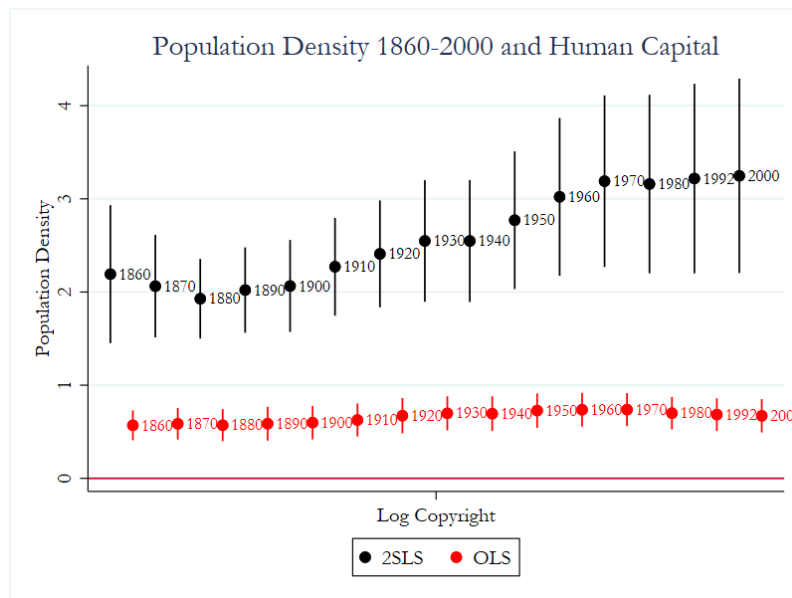


Figure 9: Point estimates with 95% confidence intervals of the effect of copyrighting in 1840-1860 on population density over 1860-2020.

Table 10: IV Regressions: Income p.c. in 2000

	(1)	(2)	(3)	(4)	(5)
Log Copyright	0.417*** (6.12)	0.421*** (5.82)	0.406*** (5.14)	0.782** (3.24)	7.988 (0.12)
% In School 1850			0.0319 (1.22)	-0.0372 (-0.79)	0.597 (0.14)
% Literate			0.149 (1.16)	-0.286 (-1.01)	2.467 (0.14)
% Urbanized				-1.696** (-3.19)	-7.826 (-0.12)
Log Population 1850					-1.432 (-0.12)
N	1512	1507	1505	1505	1504
Kleibergen-Paap F	24	28	22	13	1
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is log income per capita in 2000. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

As we can see from Figure 9, knowledge production in the pre-industrial period shows an enduring relationship with economic development. This is also evident in regressions where the outcome variable is income per-capita in 2000 as we can see in Table 10 and Figure 10. As before, when we include the full set of social controls the estimated effect of knowledge production on income per capita turns insignificant. This is due to the fact that the strength of the instrument in predicting the variable of interest is obliterated once the social controls (especially the share of foreigners and internal migrants) are introduced. While this is inconvenient, it does not necessarily invalidate our empirical strategy unless we have a strong reason to believe that these additional controls had their own independent impact on subsequent economic development. While this may be true, at least our OLS estimates always show that the variable of interest remains almost always significant even in the fully specified models.

An advantage of using title pages instead of other primary sources, such as encyclopedia subscriptions (Squicciarini & Voigtländer, 2015), to identify the knowledge elite, is that the title page can give us information on the content of publications which provides insights on the types, and not only quantities, of knowledge being produced.

Table 11: OLS Regressions: Income p.c. in 2000

	(1)	(2)	(3)	(4)	(5)
Log Copyright	0.0861*** (7.09)	0.0704*** (5.43)	0.0629*** (5.13)	0.0500*** (4.42)	0.0135 (1.04)
% In School 1850			0.0930*** (3.44)	0.0955*** (3.45)	0.0783*** (3.66)
% Literate			0.500*** (8.40)	0.518*** (9.51)	0.344*** (5.64)
% Urbanized				0.180* (2.33)	-0.180* (-2.12)
Log Population 1850					0.0122 (1.63)
N	1512	1507	1505	1505	1504
R2	0.0877	0.155	0.178	0.183	0.232
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is log income per capita in 2000. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The information available is of course limited to the title of the book, which provides a challenge when applying topic classification models as these generally perform poorly on short texts. This is why, for now, I have opted for a binary classifier separating between academic and non-academic publications. As discussed in Section 4, an Academic publication here comprises anything which serves the purpose of diffusing or advancing knowledge. As can be gleaned from Table 3 and a cursory glance at the classified primary data, a large portion of these works are Sunday School teaching manuals used by educators to teach English and other subjects to students. Another class of works consists in publications by actual academics and luminaries of the times relating to subject matters ranging from astronomy to classical literature and philosophy. Admittedly, the connection between these works and innovations fostering productivity is not always clear. While some publications in the academic category are certainly relevant to the growth of knowledge which spurred innovation and adaptation of foreign technology during the Second Industrial Revolution (see the discussion in Section 2), these are certainly a minority. However, this should not be taken to mean that the large uptake in copyrighting shown in Figures 1 and 3 is of no significance. As noted by Baten and Van Zanden (2008), the production of books, regardless of their

content, can itself indicate changes in the tastes and attitudes toward knowledge of the average consumer.

Table 12: OLS Regressions: Academic/Non-Academic

	(1) 1860	(2) 1870	(3) 1920	(4) 1970	(5) 2000
Log Science	0.00899*** (3.34)	0.00785** (2.10)	-0.00307 (-0.43)	-0.00264 (-0.45)	0.00708 (1.65)
Log Culture	0.00670*** (4.97)	0.00873*** (3.50)	0.0105*** (3.25)	-0.00254 (-0.80)	-0.00246 (-0.54)
N	1504	1504	1504	1504	1504
R2	0.295	0.195	0.203	0.221	0.130
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	yes	yes	yes	yes	yes
HC Controls	yes	yes	yes	yes	yes
Social Controls	yes	yes	yes	yes	yes

t statistics in parentheses

Dependent variable is the labor share of manufacturing 1850-2000. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

This is supported by the results shown in Table 12 which shows that, in fully specified models, non-academic works are significant predictors of the growth in the manufacturing sector while having lower point estimates. A possible explanation for this is that, while academic publications are highly clustered in certain areas (especially the Northeastern states), the production of non-academic works in the rest of the country is a marker for the general attitudes towards human capital investments. This is consistent with the evidence that counties with higher levels of copyright registrations in the pre-industrial period show higher patenting rates throughout the wave of industrialization occurring at the end of the 19th and early 20th century. Tables 14 and 13 show that patenting rates over 1860-1940 are significantly higher in counties with more copyright registrations over 1840-1860.

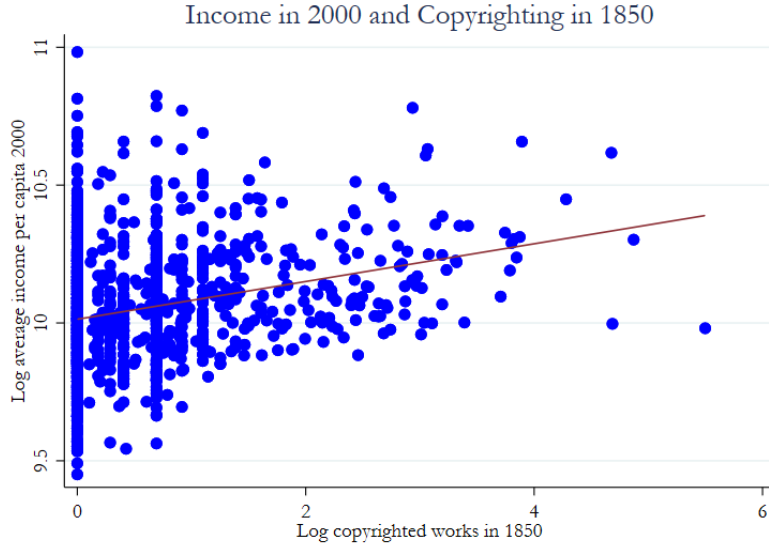


Figure 10: Scatterplot of log income per capita in 2000 and log copyright registrations in 1840-1860.

Table 13: IV Regressions: Patents 1860-1940

	(1)	(2)	(3)	(4)	(5)
Log Copyright	1.448*** (4.48)	1.416*** (4.54)	1.344*** (4.51)	2.353*** (3.59)	53.89 (0.10)
% In School 1850			0.407** (3.00)	0.220 (1.52)	3.750 (0.11)
% Literate			1.630* (2.35)	0.463 (0.42)	17.00 (0.11)
% Urbanized				-4.550** (-2.97)	-52.05 (-0.10)
Log Population 1850					-9.626 (-0.10)
N	1506	1501	1499	1499	1498
Kleibergen-Paap F	24	28	22	14	1
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is the log of patents per-capita/decade over 1860-1940. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 14: OLS Regressions: Patents 1860-1940

	(1)	(2)	(3)	(4)	(5)
Log Copyright	0.486*** (9.30)	0.441*** (12.28)	0.400*** (10.55)	0.360*** (7.47)	0.195*** (3.40)
% In School 1850			0.578*** (4.29)	0.585*** (4.33)	0.398*** (3.39)
% Literate			2.592*** (4.74)	2.650*** (4.71)	2.195*** (4.20)
% Urbanized				0.557 (1.07)	-1.098 (-1.78)
Log Population 1850					0.0856* (2.45)
N	1506	1501	1499	1499	1498
R2	0.104	0.147	0.179	0.181	0.223
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses

Dependent variable is the log of patents per-capita/decade over 1860-1940. Conley standard errors used with a 100km cutoff. For variable definitions and sources see the Data Appendix.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

6 Conclusion

This paper is the first to empirically assess the early intellectual production of the United States using copyright registration data. Constructing time series of book production at the national level reveals that authorship of books and other copyrighted materials began a sustained upward trend in 1830, which is well before the start of rapid industrialization. Assigning copyright registrations to locations using the declassified 1850 census shows that the spatial distribution of knowledge production in pre-industrial times is strongly correlated with industrial development and population density over time. Exploiting the large internal migration which occurred at this time as a result of the Westward movement, I attempt to obtain exogenous variation of “knowledge production” at the county level. IV coefficients obtained using a shift-share type instrument are statistically significant and larger than their OLS counterparts. To rationalize these results I consider a variation of the standard unified growth models developed by Galor (2011) where cultural transmission of preferences for human capital and technology can generate a long-lasting divergence in skill accumulation between regions which are ex-ante identical except for the relative scarcity of skill-using individuals. More work is needed to consider how this model can be brought to the data to perform counterfactual exercises which would illustrate how settlement patterns from centuries ago may still affect the dispersion of skills and other economic outcomes as a result of the intergenerational transmission of technology and preferences for human capital.

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7 Appendix

7.1 Human Capital and Agricultural Diversity

Fiszbein (2022) suggests that agricultural diversity played a determinant role in the US' structural change by increasing skill variety and the development of human capital. In Figure 11 I replicate Figure 7 from Fiszbein (2022) adding pre-industrial human capital, measured by the log of total copyright registrations over 1840-1860, to the IV regression. In this Figure, we plot the coefficients of 1860 agricultural diversity and pre-industrial human capital from regressions where the outcome variable is changing over time. The model we are testing is therefore the following:

$$y_{i,t} = \alpha_s + \beta_t \ln(Copyright_{1840-1860}) + \gamma Agridiv_{1860} + \Gamma'_t X_{i,1860} + \varepsilon_{i,t} \quad (5)$$

Where I instrument for both $\ln(Copyright_{1840-1860})$ and $Agridiv_{1860}$. The instrument for agricultural diversity is the same one constructed by Fiszbein (2022) so that, when I exclude $\ln(Copyright_{1840-1860})$ from the regression, Figure 11 is an exact replication of Figure 7 in Fiszbein (2022). As we can see the relationship between agricultural diversity and manufacturing disappears once we control for pre-industrial human capital and is substantially weakened in the case of population density. In the case of the labor share, the coefficients show exactly the opposite pattern that one would expect, being insignificant or close to zero prior to 1970 and positive thereafter. Repeating this exercise for pre-industrial human capital as a sanity check shows that controlling for agricultural diversity has no effect: estimates are similar to those shown in Figure 8. This shows that the relationship between economic development and pre-industrial human capital is particularly strong and, most likely, does not merely reflect a selection effect whereby knowledge producers settled in more geographically amenable locations. If this were the case, we would have expected the coefficients in panels (c) and (d) of Figure 11 to drop substantially when including Fiszbein (2022) variable capturing agricultural diversity.

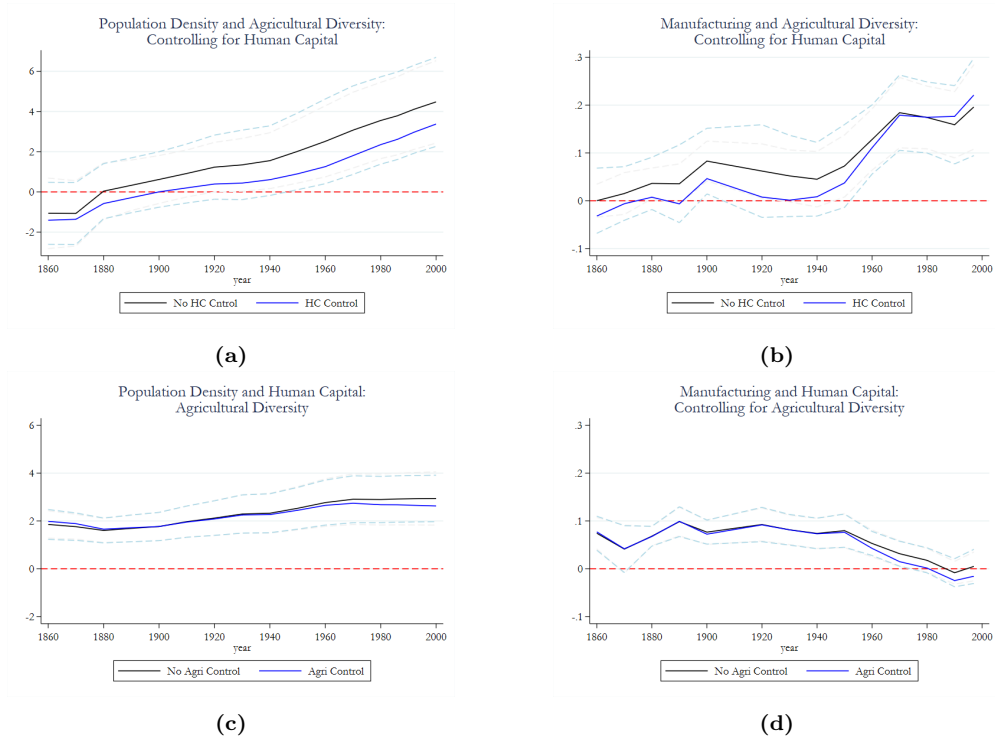


Figure 11: Replication of Fig. 7 from Fiszbein (2022). Regression results of the IV estimates of the impact of agricultural diversity, in panels (a) and (b), and pre-industrial human capital, in panels (c) and (d), on population density and manufacturing labor shares over time. Panels (a) and (c) show results for population density and panels (b) and (d) for the labor share of manufacturing.

7.2 Monte Carlo Exercise: Difference between IV and OLS Estimates

Large differences between IV and OLS estimates can be cause for concern. This is because, if the instrument is endogenous, bias in IV coefficients can be substantial even for small violations of the exclusion restriction. This especially the case for weak instruments as the formula for the IV estimator is:

$$\hat{\beta}_{iv} = \frac{cov(z, y)}{cov(z, x)} = \frac{cov(z, a + \beta x + \varepsilon)}{cov(z, x)} = \beta + \frac{cov(z, \varepsilon)}{cov(z, x)}$$

Hence, in the case of a weak instrument (which implies a low $cov(z, x)$) the bias of the IV estimator can be large even for small sample covariances of the instrument with the error term ($[cov(z, \varepsilon)] \neq 0$). It is therefore important to justify large differences between OLS and IV estimates, commenting in particular on the direction of the bias (of the OLS estimate). In our case, the OLS estimate is biased downward, by a factor of around 5-7. While this is large, I argued in Section 4 that measurement error in the variable of interest could generate this. Consider the following Monte-Carlo experiment. Suppose the d.g.p. for an outcome variable y is as follows:

$$y = a + bx + \varepsilon$$

Where x is the total amount of knowledge produced in a given geographical unit. However, part of this knowledge is unobservable (to the econometrician) which leads to measurement error. The true extent of knowledge production is divided among observable (to the econometrician) and unobservable:

$$x = d\tilde{x} + (1 - d)f$$

Where $d \in (0, 1)$ and $\tilde{x}$ is the observable knowledge and f is the unobservable component. Suppose that $\tilde{x}$ and f come from a multivariate normal distribution with covariance equal to ρ . Moreover, suppose the econometrician disposes of an instrumental variable which is equal to the sum of observable and unobservable knowledge plus an error term:

$$z = \tilde{x} + f + \epsilon$$

Where ϵ is distributed normal with mean zero and variance $\sigma_e = 1$ (increasing σ_e increases the confidence intervals around $\hat{\beta}_{iv}$ but does not introduce bias). The higher the covariance ρ and the fraction of observable knowledge d , the lower will be the downward bias of OLS estimates. Supposing that the true b is 0.7 and with a correlation of 0.9, Figure 12a shows the distribution of the OLS estimates and the mean of the IV estimates for different levels of d .

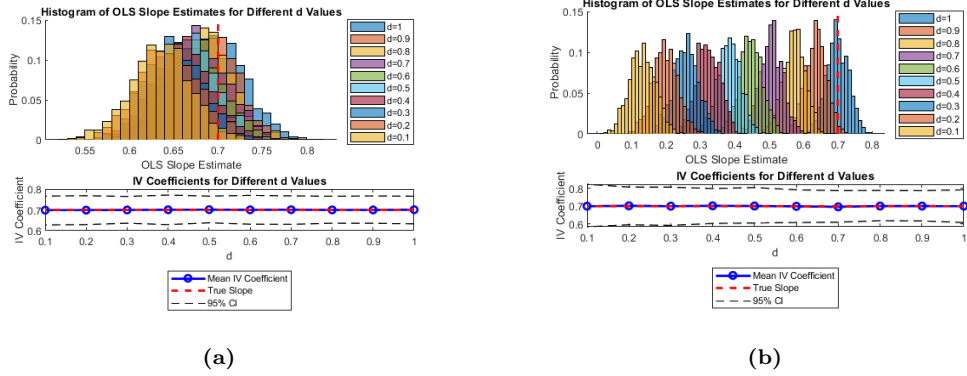


Figure 12: Monte-Carlo exercise showing the impact of measurement error on OLS and IV coefficients with $\rho = 0.9$, panel (a), and $\rho = 0.1$, panel (b), with $\sigma_e = 1$. The true parameter value is denoted by the red dotted line.

As we can see, for high correlations of observable and unobservable knowledge, the OLS estimator is informative of the true parameter value even when the fraction d of observable knowledge is small. If this correlation is lower however, perhaps due to imperfect author matching, the extent of downward bias rises rapidly. Figure 12b shows the same exercise for $\rho = 0.1$. As the fraction of observable knowledge production d is likely to be small, a large downward bias in the OLS coefficients is not unexpected. As we can see, as long as the instrumental variable is sufficiently precise (low σ_e) concerning the sum of observable and unobservable knowledge, we can recover the true parameter estimate using the IV.

7.3 Supplementary Results

Up until 1850 women were excluded from the census, which was only directed at (male) household heads. An important source of bias could ensue if a large fraction of authors prior to 1850 were women. Figure 13 shows that this was not the case, over the whole period approximately 1.3% of works were penned by women. While some female authors, such as Harriette Baker, have certainly left their mark on American literary history, when looking at the general picture female authorship appears to be markedly limited.

Turning to the match quality, Table 15 shows that over 70% of all matches are “exact matches”, which in this context means that for a given name, surname and state combination I observe only one person satisfying the occupational criteria. The list of 48 (out of 235) occupations to which I restrict the sample of potential matches is shown in Table 16 which also shows the frequency of each occupational category. It is important to note however that an “exact match” in this sense does not mean a “correct match” as I could still have made an error in extracting the author name from the title page. It is therefore important to consider how the occupational restriction of potential could drive the results. I do not show this here, but adding the log of

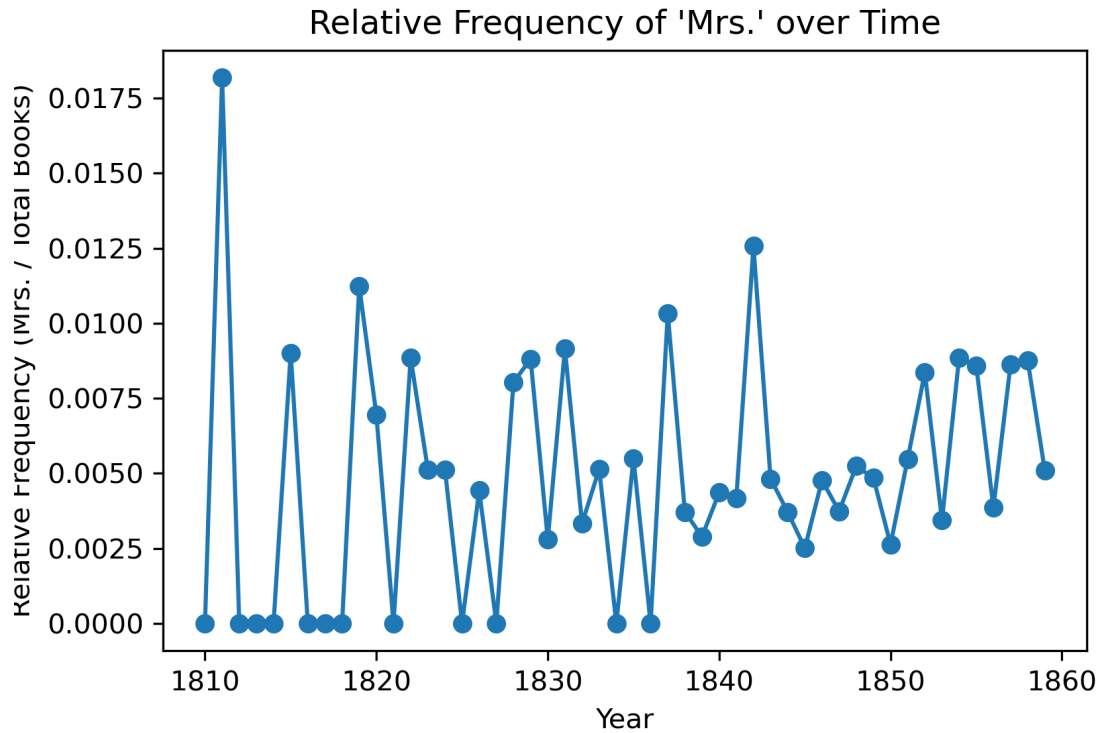


Figure 13: Relative Frequency of works containing the string “Mrs.” or “Miss” (or non-capitalized variations thereof). Only relative frequencies from years with more than 100 copyright registrations are shown.

total “potential matches”, meaning people who fulfill the occupational criteria, to the regressions in Section 6 does not qualitatively change results.

Table 15: Math Quality

Precision	Freq.	Percent
1	6927	71.69
2	990	10.25
3	585	6.055
4	364	3.767
5	255	2.639
6	180	1.863
7	133	1.377
8	88	0.911
9	90	0.931
10	50	0.517
Total	9662	100

Table showing the precision of author matches. The first column shows the number of potential matches and the following columns show the frequency and percentage of authors being matched with that level of potential matches.

Table 16: Potential Matches: Occupations

	Freq.	Percent
Accountants and auditors	1215	0.802
Architects	572	0.377
Artists and art teachers	2238	1.477
Authors	96	0.0634
Chemists	436	0.288
Clergymen	27698	18.28
College presidents and deans	79	0.0521
Agricultural sciences	1	0.000660
Chemistry	21	0.0139
Engineering	2	0.00132
Geology and geophysics	2	0.00132
Mathematics	32	0.0211
Medical sciences	11	0.00726
Physics	3	0.00198
Natural science (n.e.c.)	8	0.00528
Social sciences (n.e.c.)	12	0.00792
Nonscientific subjects	160	0.106
Subject not specified	539	0.356
Dentists	3029	1.999
Designers	113	0.0746
Draftsmen	104	0.0686
Editors and reporters	1526	1.007
Engineers, civil	812	0.536
Engineers, electrical	1	0.000660
Engineers, industrial	3	0.00198
Engineers, mechanical	49	0.0323
Engineers, mining	5	0.00330
Engineers (n.e.c.)	37	0.0244
Lawyers and judges	25129	16.58
Musicians and music teachers	3808	2.513
Nurses, professional	6	0.00396
Agricultural scientists	7	0.00462
Biological scientists	55	0.0363
Geologists and geophysicists	27	0.0178
Physicists	13	0.00858
Miscellaneous natural scientists	13	0.00858
Pharmacists	6209	4.098
Physicians and surgeons	44721	29.51

Religious workers	406	0.268
Statisticians and actuaries	4	0.00264
Teachers (n.e.c.)	31217	20.60
Technicians, medical and dental	3	0.00198
Technicians, testing	23	0.0152
Technicians (n.e.c.)	3	0.00198
Therapists and healers (n.e.c.)	370	0.244
Veterinarians	142	0.0937
Professional, technical and kindred workers (n.e.c.)	371	0.245
Opticians and lens grinders and polishers	200	0.132
Total	151531	100

Table showing the occupations which are considered potential matches and their frequency. Source: US Census of Population 1850.

Relating to the relationship between religious fractionalization and polarization and pre-industrial human capital, Figure 2 showed a, respectively, positive and negative relationship with copyright registrations. The indices of religious fractionalization and polarization are constructed following the methodology proposed by Montalvo and Reynal-Querol (2003) using data on churches at the county level. These indices are formally calculated as follows:

$$FRAC_i = 1 - \sum_{j=1}^J \left(\frac{n_{i,j}}{N_i} \right)^2$$

$$POL_i = 1 - \sum_{j=1}^J \left(\frac{0.5 - \pi_{i,j}}{0.5} \right)^2 \pi_{i,j}$$

Where i refers to counties and j to a particular religious denomination, J being the total number of denominations. N_i is the total number of churches in county i and $n_{i,j}$ is the number of churches in county i of denomination j . Intuitively, polarization is maximized when two denominations each hold 50% of the churches in a county and fractionalization increases when the number of groups increases. In Table 17 I run a series of regressions where the outcome variable is the explanatory variable used throughout the paper: the log of the total copyright registrations between 1840-1860 in each county. This exercise shows that religious polarization and fractionalization are robustly related to copyrighting activity even when controlling for the entire set of controls used throughout the paper.

Another issue which should deserve considerable attention is the demographics of authors. While the paper focuses mostly on aggregate measures of authorship at the county level, it is also interesting to ask the question: Who were the individuals who produced copyrighted works in the 19th century. Answering this question is not only important to complement the analysis carried out in the rest of the paper, but can

Table 17: OLS Regressions: Log Copyright

	(1)	(2)	(3)	(4)	(5)
FRAC	0.582*** (7.30)	0.558*** (8.05)	0.523*** (7.92)	0.523*** (7.92)	0.268*** (5.59)
POL	-0.505*** (-6.38)	-0.476*** (-6.73)	-0.445*** (-6.23)	-0.445*** (-6.23)	-0.215*** (-5.14)
N	1511	1506	1505	1505	1504
State FE	yes	yes	yes	yes	yes
Lat/Lon Polynomial	yes	yes	yes	yes	yes
Geography Controls	no	yes	yes	yes	yes
HC Controls	no	no	yes	yes	yes
Social Controls	no	no	no	no	yes

t statistics in parentheses.

Dependent variable is log of total copyright registrations over 1840-1860. Conley standard errors used with a 100km cutoff. Definitions of variables and sources are contained in the Data Appendix.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

also assuage concerns regarding the accuracy of the matching process. If the group of authors is a random subset of the population (due to purely spurious matching) the we should not observe systematic differences between authors and non-authors. Table 18 shows that this is not the case, indicating that the author names used in the matching process are adding a valuable source of information. Authors were richer (in terms of property value), older, more foreign and less white than other people in the potential match group (defined as individuals belonging to one of the occupations in Table 16).

Table 18: Author Descriptive Statistics

	Author		Non-Author		Diff
	Mean	SD	Mean	SD	t
White	0.990	0.098	0.995	0.073	(3.105)
Real estate value	2395.759	11704.767	1679.085	10034.188	(-4.343)
Foreign	0.158	0.364	0.118	0.323	(-7.658)
Age	38.810	12.924	35.834	12.731	(-16.271)
Observations	5162		146369		151531

Descriptive statistics for authors and non-authors. The non-authors reference category is composed of individuals belonging to one of the occupations in Table 16 who do not figure in the copyright registrations. The variables “Foreign” and “White” take on value 1 if an individual is foreign or white and 0 otherwise. Real estate value is defined in 1850 dollars. Source: US Census of Population 1850.

I do not show the distribution of occupations for authors as it is very similar to that presented in Table 16. This effectively means that the differences between authors and non-authors reported in Table 18 is not due to occupational selection.

Table 19: Controls

Variable	Geography	Social	Human Capital
Potential Yields	✓		
Average Rainfall	✓		
Temperature	✓		
Distance to Oceans or Great Lakes	✓		
Terrain Ruggedness	✓		
Log of Population		✓	
Urbanization		✓	
Fraction of Foreigners		✓	
Internal Migrants		✓	
Population Density		✓	
Literacy Rate			✓
School Enrollment Rate			✓

7.4 Data Appendix

In this paper I use three main data sources. First, the data on copyright registrations is obtained from the website of the Library of Congress (see Section 4 for more information). Second, data relating to all controls and outcome variables (excepting urbanization literacy and school enrollment rates) comes from Fiszbein (2022). Lastly, data on religious denominations, city growth, literacy and school enrollment rates comes from the ICPSR (Haines et al., 2010). I also check whether the data in Fiszbein (2022) is consistent with the data from the ICPSR where the two provide the same indicators and find this to be the case. As discussed in Section 6, there are three types of controls relating to: (i) geography, (ii) human capital and other socially determined variables. It is important to distinguish between the first and remaining groups as the latter will be potentially endogenous to human activity and hence may inadvertently capture a channel through which the variable of interest affects the dependent variable. To limit this concern, these controls are always measured at baseline (1850). Table 19 shows the variables falling in each group of controls.

As far as the classification of academic/non-academic works, table 20 contains the list of keywords used to classify a work as academic.

The classification procedure is done as described in Section 4.

Table 20: Keywords

Keywords
textbook
reader
instruction
principal
grammar
school
Sundayschool
Sabbath school
manual
teacher
university
professor